

TADOULE LAKE, MB, Canada

WMO: 713340

Lat: 58.709N

Lon: 98.501W

Elev: 860

StdP: 14.25

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.6	-34.2	-45.5	0.4	-37.5	-41.6	0.5	-34.0	18.4	-6.4	16.9	-9.5	6.0	300	0.780

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	79.1	60.5	75.6	59.2	72.0	57.9	63.6	73.9	61.5	70.9	59.6	68.6	8.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.6	78.7	67.2	57.6	73.0	65.0	55.6	67.9	63.2	29.5	73.6	27.9	70.7	26.5	69.0	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	15.9	14.2	-39.1	85.8	0.5	2.1	-39.5	87.3	-39.8	88.5	-40.1	89.6	-40.5	91.1
			-39.3	68.5	0.5	1.6	-39.7	69.6	-40.0	70.5	-40.3	71.4	-40.7	72.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	-13.2	-11.6	1.1	19.2	35.8	52.5	60.0	56.1	45.2	29.2	6.5	-8.9		
	DBStd	28.53	13.06	11.74	13.91	13.98	10.45	8.66	6.28	6.18	7.69	8.61	13.80	13.54		
	HDD50	10628	1958	1724	1517	924	453	74	5	14	185	645	1304	1825		
	HDD65	15419	2423	2144	1982	1374	904	382	178	285	594	1109	1754	2290		
	CDD50	723	0	0	0	0	14	149	315	203	41	1	0	0		
	CDD65	39	0	0	0	0	0	7	23	9	0	0	0	0		
	CDH74	463	0	0	0	0	10	121	224	102	6	0	0	0		
	CDH80	61	0	0	0	0	1	14	33	13	0	0	0	0		
	WSAvg	7.2	7.0	7.1	7.2	7.3	7.4	7.4	7.1	7.2	7.9	7.8	7.0	6.6		
	PrecAvg	16.9	0.6	0.5	0.8	0.7	1.2	2.0	2.9	2.6	2.4	1.7	0.8	0.6		
	PrecMax	25.9	1.4	1.3	2.2	1.5	2.5	3.3	5.6	5.2	4.4	3.2	1.9	1.3		
(15) Precipitation	PrecMin	12.8	0.2	0.1	0.2	0.2	0.2	0.7	1.5	1.0	0.8	0.6	0.2	0.2		
	PrecStd	2.7	0.3	0.3	0.5	0.4	0.6	0.7	1.0	1.0	0.8	0.7	0.4	0.3		
	DB	26.7	25.0	43.1	55.8	74.5	81.9	84.3	81.8	72.4	57.5	39.9	25.2			
	MCWB	26.1	22.4	35.5	44.4	54.8	59.0	65.0	64.6	58.1	47.4	35.7	23.6			
	2% DB	20.0	17.2	36.6	48.1	66.4	77.7	79.4	76.8	65.3	49.4	33.7	22.0			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	19.5	16.0	31.4	39.3	50.3	57.8	61.0	61.7	54.7	43.6	32.1	21.3			
	5% DB	13.7	12.9	30.7	43.1	60.0	73.5	76.2	71.8	60.9	44.3	29.7	17.2			
	MCWB	13.1	11.7	27.7	36.2	47.9	56.0	60.0	59.7	52.6	41.0	28.6	16.6			
	10% DB	7.0	8.1	24.0	38.9	53.7	69.3	72.7	67.7	56.8	40.3	26.4	11.7			
	MCWB	6.4	7.1	21.8	33.7	44.3	54.9	58.9	57.8	50.3	38.0	25.5	11.0			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	26.3	22.9	35.7	45.6	56.7	62.5	67.1	66.1	59.3	48.0	36.1	24.1			
	MCDB	26.6	25.3	42.5	55.0	70.8	74.6	78.2	79.0	70.1	53.7	38.8	24.8			
	2% WB	19.5	15.8	31.6	39.9	51.8	59.8	64.2	63.2	56.3	44.3	32.2	21.3			
	MCDB	20.0	17.1	35.8	47.2	65.8	72.3	74.2	73.8	63.1	49.1	33.6	22.1			
	5% WB	13.0	11.7	27.5	36.5	48.3	57.6	62.3	60.9	53.6	40.9	28.7	16.5			
(33) Mean Daily Temperature Range	MCDB	13.6	12.9	30.9	42.2	59.9	69.9	71.6	69.6	59.5	44.0	29.6	17.0			
	10% WB	6.3	7.1	21.9	34.0	44.7	55.7	60.5	58.7	51.1	38.0	25.4	11.0			
	MCDB	6.9	8.2	24.0	38.6	52.8	68.0	69.1	65.6	56.2	40.2	26.4	11.6			
	5% DB	15.5	18.5	22.6	21.8	19.7	21.0	18.3	16.4	13.4	9.7	13.7	14.1			
	MCDBR	20.2	22.5	27.6	23.3	31.2	28.4	24.4	23.9	19.8	15.9	13.5	16.8			
(36) Clear-Sky Solar Irradiance	MCWBR	20.1	21.2	23.0	15.9	18.0	12.8	10.5	12.0	11.2	11.0	12.5	16.4			
	5% WB	20.1	22.3	26.1	21.9	29.5	24.6	20.1	21.6	18.2	13.9	13.0	16.7			
	MCWBR	19.9	21.1	21.9	15.3	17.5	12.4	10.2	11.5	11.3	10.0	12.1	16.5			
	taub	0.213	0.234	0.257	0.343	0.333	0.342	0.386	0.359	0.311	0.273	0.236	0.185			
	taud	2.145	2.269	2.265	2.058	2.327	2.433	2.354	2.436	2.548	2.471	2.119	2.065			
(42) All-Sky Solar Radiation	Ebn at Noon	205	262	286	271	285	283	266	266	266	245	184	175			
	Edh at Noon	14	21	29	43	36	33	35	30	23	18	15	10			
	RadAvg	190	512	1063	1643	1849	1829	1659	1274	810	383	182	112			
(45)	RadStd	14	36	71	133	148	181	115	102	76	41	22	10			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page